MCO-1D ORNITHOPTER

Bill of Materials — Revision 16 — July 2026

Complete BOM reflecting the full programme as of Revision 16, including the wing morphing programme (H-33 through H-36), pitot airspeed sensor, load cell commissioning equipment, and all new tooling and diagnostic software.

Aircraft	MCO-1D metachronal cascade ornithopter
MTOW	3.448 kg (7.60 lb) — measure actual after build
Wings	8 — four tandem pairs W1 (front) to W4 (rear)
Motors	8 × GoolRC 5010-14 360 KV — one per wing, fully independent
ODrive boards	4 × v3.6 — axis0 = port wing, axis1 = starboard per pair
Pitch servos	8 × feedback-capable digital servo (AoA control)
Sweep servos	8 × feedback-capable high-torque servo (fore-aft + stow)
Pitch ADC	MCP3208 on SPI1 CE1 — actual AoA per wing at 50 Hz
Sweep ADC	MCP3208 on SPI1 CE0 — actual sweep per wing at 50 Hz
Stroke sensing	ODrive pos_estimate CAN — no extra hardware
Pitot sensor	Pixhawk-compatible pitot tube — airspeed for all morphing
Wave Library	48,001 lines — 33 hypotheses, 180 variants, 148 GS commands
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1. AIRFRAME AND STRUCTURE

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	set	Main airframe / fuselage	Must mount 8 sector wheel axles, 8 motor brackets, 16 servo brackets, 4 ODrive boards, Pi 4, all sensors, pitot tube. Rigid: no flex between motor mounts under full-thrust vibration.	Custom
8	pcs	Wing panels — S1223 aerofoil, chord 165 mm (measure actual)	Independent per wing. Must sustain 96° amplitude at 3.0 Hz. W1=772, W2=749, W3=735, W4=712 mm semi-span (estimates — measure actual). Sweep servo must be able to rotate wing to ~85-95° aft of neutral for H-33 State C (fuselage stow) operation.	Custom
8	pcs	Sector wheel assemblies — 100° mechanical arc	One per wing. Hard stop at 100°. Software limits to 96°. Must support full arc smoothly under aerodynamic and inertial loads.	Custom
8	sets	Rope drive — one per wing, fully independent	NO shared rope between port and starboard. Motor drum → rope → sector wheel. Inspect before every flight. Current asymmetry between port/stbd at same pair = early rope wear indicator.	Cord
8	pcs	Wing pivot, AoA linkage, and sweep linkage hardware	Pitch servo controls AoA ±22°. Sweep servo controls fore-aft ±30-40° for dynamic sweep (H-34) and up to 85-95° aft for fuselage stow (H-33 State C). Measure and record actual sweep limits before TC-01.	Hardware
1	set	Pitot tube mounting hardware	Mount pitot facing directly into clean airflow, away from wing downwash and prop wash. Forward fuselage nose or forward boom. Connect to Pixhawk airspeed port. ALL morphing hypotheses (H-33/34/35/36) use pitot airspeed — GPS ground speed is not used for any aerodynamic decision.	Custom

2. MOTORS — 8 INDEPENDENT ASSEMBLIES

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
8	pcs	GoolRC 5010-14 Brushless Motor — 360 KV	12N14P, 7 pole pairs (verified). One motor per wing. Port and starboard on same pair driven by axis0/axis1 of one ODrive. Confirm 5010-14 not 5010-6.	~$25
8	pcs	Motor output drum for rope drive	7:1 gear ratio: motor 7 turns → sector wheel 1 turn. Verify ratio: rotate motor shaft 7 times, confirm sector turns once.	Custom

8	sets	Motor mount hardware with vibration isolation	Medium thread lock (Loctite 243) on all fasteners. Vibration isolation pads between motor base and frame. Re-torque after first 10 flights.	Hardware

3. ODRIVE MOTOR CONTROLLERS — 4 BOARDS × 2 AXES

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
4	pcs	ODrive v3.6 motor controller	Board 0 node 0: axis0=W1p axis1=W1s. Board 1 node 1: axis0=W2p axis1=W2s. Board 2 node 2: axis0=W3p axis1=W3s. Board 3 node 3: axis0=W4p axis1=W4s. vel_limit=150 rad/s (set by mco_odrive_setup.py). pole_pairs=7. MOTOR_TYPE_HIGH_CURRENT. Velocity control mode.	~$130
1	pcs	MCP2515 or MCP2518FD SPI-to-CAN adapter	SPI0 CE0 to Pi. MCP2515: 8 MHz oscillator, interrupt GPIO25. Daisy-chain: Pi → ODrive 0 → 1 → 2 → 3.	~$5
2	pcs	CAN bus 120 Ω termination resistors	One at Pi-adapter end, one at ODrive 3 end. Between CAN-H and CAN-L. Both required.	<$1
2+	m	CAN bus twisted pair cable	Twisted CAN-H and CAN-L. Daisy-chain all 4 boards.	Per m

4. PITCH SERVOS — ANGLE OF ATTACK — 8 WINGS

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
8	pcs	Digital servo with analog feedback pot output — pitch (AoA)	MUST have feedback potentiometer output pin (3rd wire beyond power/signal). 333 Hz PWM capable. 7.4 V supply from dedicated servo LiPo. GPIO PWM: W1p=18, W1s=23, W2p=24, W2s=25, W3p=12, W3s=16, W4p=20, W4s=21. Feedback → 1kΩ resistor → MCP3208 #1 (SPI1 CE1).	~$15
8	wires	Pitch feedback signal wires + 1 kΩ resistors	W1p→CH0, W1s→CH1, W2p→CH2, W2s→CH3, W3p→CH4, W3s→CH5, W4p→CH6, W4s→CH7. Solder 1 kΩ resistor in-line at ADC input end.	<$2

5. SWEEP SERVOS — FORE-AFT POSITION AND STOW — 8 WINGS

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
8	pcs	High-torque digital servo with analog feedback pot — sweep	MUST have feedback pot output. 25 kg·cm+ torque. 7.4 V. Must reach ~85-95° aft of neutral for H-33 State C (fuselage stow). Must execute ±20-40° excursion within 200 ms for H-34 dynamic sweep. GPIO PWM: W1p=4, W1s=17, W2p=27, W2s=22, W3p=5, W3s=6, W4p=13, W4s=19. Feedback → 1kΩ → MCP3208 #2 (SPI1 CE0).	~$12
8	wires	Sweep feedback signal wires + 1 kΩ resistors	Same channel mapping as pitch: W1p→CH0 ... W4s→CH7.	<$2

6. PER-WING SENSING — MCP3208 ADC CHIPS

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
2	pcs	MCP3208-CI/P 8-channel 12-bit SPI ADC	ADC #1 PITCH: SPI1 CE1 (GPIO7). ADC #2 SWEEP: SPI1 CE0 (GPIO8). Both share SPI1 bus: CLK=GPIO11, MOSI=GPIO10, MISO=GPIO9. VREF=3.3V, VDD=3.3V. GND to Pi GND. Resolution: 12-bit → 0.8 mV/step → ~0.044°/step. /boot/config.txt: dtoverlay=spi1-2cs	~$3
16	pcs	1 kΩ resistors 0.25W (8 pitch + 8 sweep feedback lines)	Series protection on each ADC input. One per servo feedback wire.	<$1

7. FLIGHT COMPUTER — RASPBERRY PI 4

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	pcs	Raspberry Pi 4 Model B — 4 GB RAM minimum	4 GB required. Wave Library + aero state engine + discovery registry exceed 500 MB during extended Discovery Mode. Tick rate 50 Hz. Mean tick ~13 ms on Pi 4 (1.5× headroom). Vibration isolate mount — use neoprene pads.	~$55
1	pcs	MicroSD card — 64 GB high-endurance Class 10 / A1	High-endurance rated. Logs grow ~50 MB/hr at 50 Hz. 64 GB = 1,280 flight hours. Never store logs on tmpfs. Never share power rail with motors.	~$12
1	pcs	Pi heatsink + active cooling case	Motor vibration environment. Active cooling fan recommended for summer operations and extended Discovery Mode.	~$10

8. SENSORS

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	pcs	MPU-6050 IMU — I2C 0x68	6-DOF at 100 Hz. Roll, pitch, yaw rates + accelerations. Stall detection Tier 1 (vibration correlation) and Tier 4 (lift waveform). I2C1: SDA=GPIO2, SCL=GPIO3. Vibration isolate.	~$3
1	pcs	BMP280 barometer — I2C 0x76	Altitude above launch. Active above 3 m where lidar loses accuracy. Same I2C1 bus as IMU.	~$3
1	pcs	TF-Luna lidar rangefinder — 50 Hz	Height 0.1–8 m above ground or water. Primary for landing. Over water: reads wave surface. Mount facing straight down, clear of wing arc.	~$20
1	pcs	DS18B20 waterproof OAT sensor	Outside air temperature for density altitude. GPIO4 One-Wire. 4.7 kΩ pull-up to 3.3 V. /boot/config.txt: dtoverlay=w1-gpio,gpiopin=4	~$3
1	pcs	Pitot tube — Pixhawk-compatible airspeed sensor	CRITICAL for all morphing hypotheses H-33/34/35/36. V_ms = pitot airspeed (msg.airspeed from Pixhawk VFR_HUD). GPS ground speed is NEVER used for aerodynamic decisions. Standard Pixhawk MS4525DO digital airspeed sensor recommended. Mount in clean undisturbed air forward of all wings.	~$20
1	pcs	HX711 ADC + load cell 50 N minimum — BENCH USE ONLY	Static thrust during TC-01 and TC-02. Not flown. HX711 DT=GPIO26, SCK=GPIO7. Read by mco_loadcell.py. Calibrate before TC-01 with known mass. Inline between aircraft and post fixture.	~$12

9. RADIO AND NAVIGATION

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	set	RC transmitter + SBUS receiver — 4 channels minimum	CH1=Roll, CH2=Pitch, CH3=Lever, CH4=Yaw. Verify with mco_rc_verify.py before TC-01. Wrong mapping = loss of control. SBUS → Pi UART (GPIO14 TX, GPIO15 RX). Bind before first power-on.	Per system
1	pcs	Pixhawk flight controller	SENSOR BRIDGE only — does not run flight control. Provides GPS, compass, extra IMU, and PITOT AIRSPEED via MAVLink. Connect Pixhawk TELEM2 → Pi UART at 921,600 baud. GPS module required for Discovery Mode positioning.	~$40

1	pcs	GPS module for Pixhawk	Mount as far from motors as possible. Used for ground speed (navigation) and Discovery Mode racetrack. NOTE: pitot tube provides aerodynamic airspeed — GPS is not used for any aerodynamic calculation.	~$20
1	pcs	Touchscreen display 3.5 inch+	Runs mco_panel.py via mco-panel.service. 4 buttons: PRE-FLIGHT, ARM, DISCOVERY, POST-FLIGHT. 6 margin bars: altitude, stall, frequency, amplitude, gust, combined. Bars: amber=50%, red=25%.	~$25

10. POWER SYSTEM

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	pcs	Main LiPo — 4S, 4,000 mAh minimum, 30 C discharge	5,000–6,000 mAh recommended for Discovery Mode. 4 × ODrive boards + 8 motors. Monitor via ODrive telemetry. LOW at 30%, CRITICAL at 20%.	Per spec
1	pcs	Servo LiPo — 2S, 1,500 mAh, 7.4 V	DEDICATED supply for all 16 servos. Keeps motor PWM noise off servo rail. Sweep servos under dynamic load (H-34) draw more current than static hold — size battery accordingly.	Per spec
1	pcs	BEC 5 V / 3 A isolated — Pi supply	ISOLATED from motor supply. Motor PWM noise on shared rail corrupts SD card writes and knowledge registry.	~$8
1	pcs	30 A blade fuse + holder — main battery lead	Closest to battery.	~$3
1	pcs	5 A blade fuse + holder — Pi BEC output		~$2

11. COMMISSIONING EQUIPMENT

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	set	Static test post fixture	Rigid frame. Aircraft bolts to post. Load cell inline between aircraft attachment point and post. Must withstand >40 N without movement. Used for TC-01 (bench) and TC-02 (captive hover). Same fixture for both — no reconfiguration needed.	Custom
1	pcs	Load cell — 50 N capacity minimum, compression type	Inline between aircraft and post. Reads DECREASING force as flapping frequency increases. Reads ZERO when aircraft generates lift equal to its own weight. That zero point IS the hover frequency. Read by mco_loadcell.py via HX711 (GPIO26 DT, GPIO7 SCK).	~$15
1	pcs	Calibration mass — 1.000 kg accurately known	For mco_loadcell.py --calibrate before TC-01. Use a certified 1 kg mass or known accurate object.	~$10
1	pcs	Truck or flatbed vehicle with open bed — TC-03	Provides forward airspeed for TC-03. Aircraft mounted to truck bed. Truck drives at commanded speeds (software shows target on GS). Minimum: 3–15 m/s (6–29 kt) available speed range. Driver and GS operator communicate by radio.	Borrowable
1	pcs	IR thermometer or thermal camera	Motor temperature during TC-01-11 (frequency ceiling). Target: below 60°C at operating frequency after 60 seconds. Check all 8 motors individually.	~$20
1	pcs	Laptop with Python 3.8+ and WiFi	Ground station. Runs mco_ground_station.py on port 5760. Also runs mco_log_analysis.py after each TC session. Second operator required during all TC operations.	Per availability
1	pcs	Tape measure — 2 m minimum	For mco_build_measurements.py. Measure all 8 wing spans, chord, fuselage length, CG, sweep range, sector arc.	~$5
1	pcs	Kitchen scale — 5 kg capacity, 1 g resolution	For mco_build_measurements.py MTOW measurement. Weigh aircraft with flight battery installed.	~$15
1	pcs	Protractor or digital angle gauge	For mco_build_measurements.py sector arc and sweep angle measurement. Also for TC-01 servo position visual confirmation.	~$10
1	set	Field safety kit	LiPo-rated fire extinguisher. First aid. Safety glasses. Hearing protection. 30 m clear zone around aircraft during TC-02.	Per site

12. CONSUMABLES

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	roll	Thread lock medium (Loctite 243 blue)	All motor mount, sector wheel axle, and ODrive mount fasteners.	Per roll
1	can	Conformal coating — water ops only	Pi, ODrives, ADC boards, HX711. Thin coats, avoid connectors.	Per can
2	pcs	Spare rope drive sections	Pre-cut and pre-tensioned. Rope is primary wear item. Rising motor current before failure is the warning sign.	Per length
10	pcs	MicroSD write-protected labels	Label SD card with aircraft serial number and last backup date.	<$1

13. SOFTWARE — ALL FREE / OPEN SOURCE

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	file	mco_wave_library.py — 48,001 lines	50 Hz tick stack. 33 hypotheses H-01 to H-36, 180 variants. 148 GS commands. Full optimizer with morphing speed scheduler. H-33: wing stow/stop (18v). H-34: dynamic sweep (16v). H-35: thrust/lift separation (14v). H-36: loiter/ferry (8v).	No cost
1	file	mco_main.py — 405 lines	50 Hz flight loop entry point.	No cost
1	file	mco_odrive_interface.py — 253 lines	4-board 8-axis CAN interface. axis0 and axis1 per board.	No cost
1	file	mco_loadcell.py — 527 lines	HX711 load cell reader. GPIO26 DT, GPIO7 SCK. Calibration, taring, continuous monitoring, thrust sweep CLI.	No cost
1	file	mco_build_measurements.py — 328 lines	Guided 15-measurement physical survey of aircraft. Patches real values into wave library.	No cost
1	file	mco_preflight_checks.py — 573 lines	14-check pre-flight system. Battery interlock (80% minimum for TC). Arming blocked if any FAIL. Integrates into mco_main.py.	No cost
1	file	mco_rc_verify.py — 324 lines	RC channel mapping verification. Confirms CH1-4 assignment. Lever direction check. Lever-safe arming interlock.	No cost
1	file	mco_vibration_check.py — 367 lines	IMU vibration sweep 0.5–3.0 Hz. SPI reliability under vibration. Tick budget measurement. Flags damping problems.	No cost
1	file	mco_test_cards.py — 1,665 lines	TC-01 (16 steps), TC-02 (8 steps), TC-03 (6 steps), TC-04 (7 steps). TC-01/02/03 fully automatic. TC-04 guided. Results persist to JSON. Load cell integrated into TC-01/02.	No cost
1	file	mco_log_analysis.py — 462 lines	Post-flight log analysis. Per-wing AoA stats, anomaly detection, current asymmetry, thrust curve extraction, hover freq extraction. ASCII time-series plots.	No cost
1	file	mco_ground_station.py — 660 lines	Laptop ground station. Sensing display, morphing speed report, loiter/ferry report. 148 commands.	No cost
8+	files	Supporting files	mco_odrive_setup.py, mco_preflight.py, mco_logger.py, mco_mavlink.py, mco_gs_server.py, mco_panel.py, mco_servo_interface.py, mco_hypothesis_tool.py, mco_sync.py, mco_can_setup.sh, mco_install.sh.	No cost

1	file	Raspberry Pi OS Lite 64-bit	raspberrypi.com/software. Lite (no desktop) for maximum CPU headroom.	No cost

COMPLETE HARDWARE COUNT

Motors	8 (one per wing, all independent)
ODrive boards	4 (axis0 = port, axis1 = starboard per pair)
Pitch servos	8 (feedback potentiometer output required)
Sweep servos	8 (feedback pot required; must reach fuselage-stow position)
Total servos	16
Pitch ADC	1 × MCP3208 SPI1 CE1
Sweep ADC	1 × MCP3208 SPI1 CE0
1 kΩ resistors	16 (one per servo feedback line)
CAN bus	4 ODrives daisy-chained, 120 Ω at each end
Pitot tube	1 — Pixhawk-compatible, REQUIRED for H-33/34/35/36
Load cell	1 × 50 N minimum + HX711 — bench only, read by mco_loadcell.py
Flight computer	Raspberry Pi 4, 4 GB RAM
Wave Library	48,001 lines Python
Hypotheses	33 active (H-01 to H-36), 180 variants
GS commands	148
Discovery modes	12
TC-01	Bench test — 16 steps, fully automated (6 semi-auto)
TC-02	Captive hover on same post — 8 steps, fully automatic
TC-03	Truck-mount forward flight — 6 steps, fully automatic
TC-04	First free flight — 7 steps, guided automatic
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■ WARNING: Every aerodynamic decision in H-33/34/35/36 uses PITOT AIRSPEED from the Pixhawk pitot tube (V_ms = msg.airspeed). GPS ground speed is never used for aerodynamic calculations.

✓ Sweep servos must reach approximately 85-95° aft of neutral for H-33 State C (full fuselage stow). Measure the actual sweep range with mco_build_measurements.py before TC-01.

✓ Load cell and HX711 are bench-only commissioning tools. Not flown. Required for TC-01 thrust curve and TC-02 hover frequency.